Dylan Thornburg ECEN 133 Prelab 9

1. Use two familiar short one dimensional FIR filters :
   - $h1 = [1,2,1]$ is a second order low pass filter with even symmetry. It has two zeros at $z = -1$. $H(z) = \frac{z^2+2z+1}{z^2} = \frac{(z+1)^2}{z^2}$.
   - $h2 = [1,0,-1]$ is a central difference filter. It is a second order band pass filter with odd symmetry, and its two zeros are at $z = 1$ and $z = -1$. $H(z) = \frac{z^2-1}{z^2} = \frac{(z+1)(z-1)}{z^2}$.

   On paper, find the filter coefficients of the one-dimensional filter $h3 = h1 * h1$ where * means convolution. Where are its zeros? What kind of filter is it?

   On paper, find the coefficients of two-dimensional filters using the outer products (matrix multiplication of a column vector and a row vector) of two one-dimensional filters. Here the one-dimensional filters are defined as row vectors. What are the matrix dimensions of each two-dimensional filter matrix?

   - $hA = h1' \cdot h1$ and $hB = h2' \cdot h1$ and $hC = h3' \cdot h3$

1. $h_1 = [1,2,1]$ $\frac{(z+1)^2}{z^2} = H(z)$ $h_2 = [1,0,-1]$ $H(z) = \frac{(z+1)(z-1)}{z^2}$

a.

$h3 = (z^2+2z+1)(z^2+2z+1) = z^4 + 4z^3 + 6z^2 + 4z + 1 = [1,4,6,4,1]$

zeros: 4 zeros at -1 at $w=0$, output $=16$ Low pass filter

at $w=\pi$, output $=0$

b.

$hA = h1' \cdot h1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} [121] = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ 3x3

$hB = h2' \cdot h1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} [121] = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$ 3x3

$hC = h3' \cdot h3 = \begin{bmatrix} 1 \\ 4 \\ 6 \\ 4 \\ 1 \end{bmatrix} [14641] = \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$ 5x5

2. Create the six filters from above with MATLAB. Remember that in MATLAB you will use * for matrix multiplication and the *conv* function for one dimensional convolution. Compare the size of the filters and the values with your earlier computations. Submit your MATLAB instructions and a MATLAB display of the values.
3. Create a test array and display it with the MATLAB function *mesh* using

   A1 = zeros(16);
   A1(5:11,5:11) = ones(7);
   mesh(A1)

   You should see a 3D perspective plot of the surface A1, which has a height of 1 in its central area and a frame of zeros near the edges.
4. Filter the test array using two-dimensional convolution with *y = conv2(hC, A1);* and display the result with mesh.
5. Read a color image file and display it as an image.
   - Read the file with *i1 = imread('4.1.05.tiff');* What is the size of the image? (Use the MATLAB *size* function.)
   - Display the image with *imshow(i1);*

```
h1 = [1, 2, 1];
h2 = [1, 0, -1];
h3 = conv(h1, h1); % h3 = [1, 4, 6, 4, 1]

hA = h1' * h1; % 3x3
hB = h2' * h1; % 3x3
hC = h3' * h3; % 5x5

disp('hA ='); disp(hA);
```

```
hA =
     1     2     1
     2     4     2
     1     2     1
```

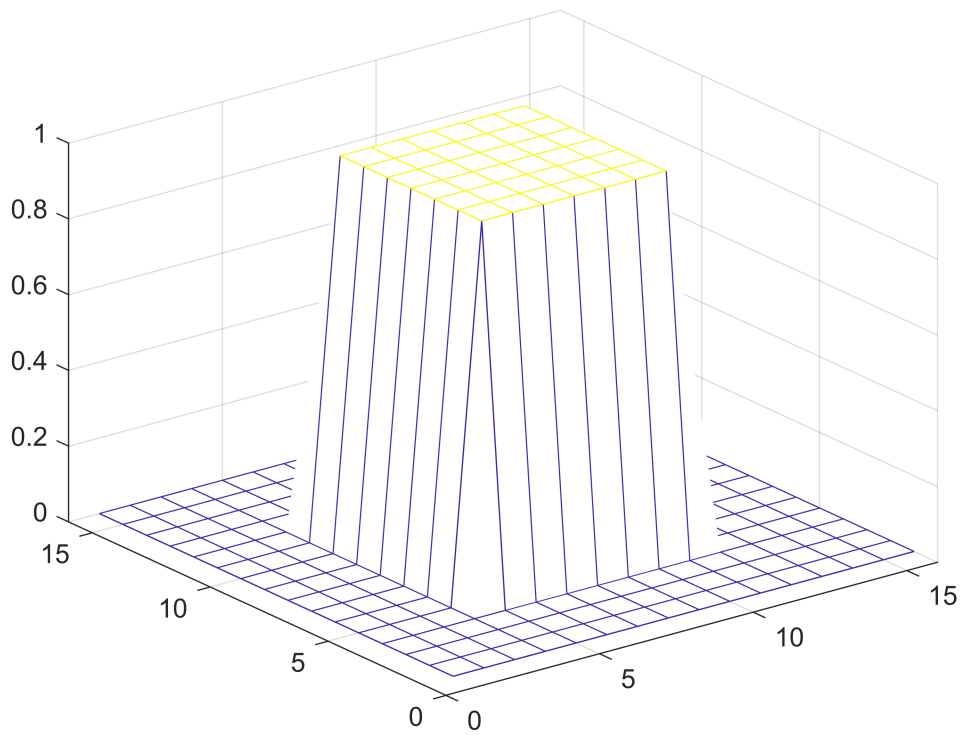
```
disp('hB ='); disp(hB);
```

```
hB =
     1     2     1
     0     0     0
    -1    -2    -1
```

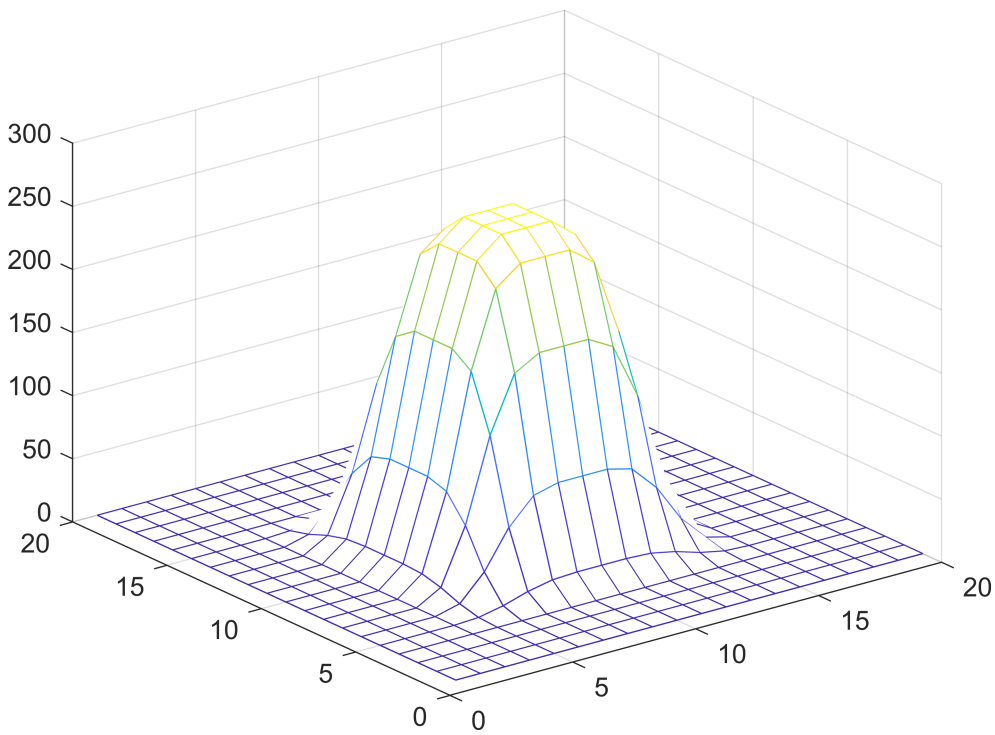
```
disp('hC ='); disp(hC);
```

```
hC =
     1     4     6     4     1
     4    16    24    16     4
     6    24    36    24     6
     4    16    24    16     4
     1     4     6     4     1
```

```
% Test array A1
A1 = zeros(16);
A1(5:11, 5:11) = ones(7);
figure;
mesh(A1);
```




```
% Filter A1 with hC
y = conv2(hC, A1);
figure;
mesh(y);
```




```
i1 = imread('4.1.05.tiff');
size(i1) % image has depth of 3 for some reason
```

```
ans = 1×3
   256   256     3
```

```
figure;
imshow(i1);
```